

M. M. Institute of Computer Technology & Business Management (MCA), M.M. (Deemed To Be University), Mullana					Course Coverage Plan	
Session: 2025- 2026 (Even)			Class/Semester: BCA – 6 th Semester			
Subject Code: BCA-606			Subject Name: Machine Learning			
Lecture: 3			Tutorial: 0			
Max. Marks Internal Practical : 60			Max. Marks Practical = 100		Credit 3.0	
Section	Lecture No.	Date	Lecture Plan	Referred Book	Mode of Delivery (Regular/ICT)	Method of ICT
A	1	7.01.2026-12.01.2026	Introduction to Machine Learning (ML): History and features of ML	[1]	Regular	NA
	2	14.01.2026-16.01.2026	working of ML, Classification of ML	[2]	Regular	NA
	3	19.01.2026-21.01.2026	Data Pre-Processing: Data Frame Basics, CSV File	[1]	Regular	NA
	4	23.01.2026	Libraries for Pre-processing	[1][2]	Regular	NA
	5	23.01.2026-	Handling Missing data	[1][2]	Regular	NA
	6	28.01.2026	Encoding Categorical data	[1][2]	Regular	NA
	7	28.01.2026	Feature Scaling, Handling Time Series data.	[1][2]	Regular	NA
	8	30.01.2026-	Feature Extraction: Overview of Feature Selection	[1][2]	Regular	NA
	9	02.02.2026	Feature Extraction Techniques; Data Transformation,	[1][2]	Regular	NA
	10	02.02.2026	Data Normalization.	[1][2]	ICT	PPT
B	11	04.02.2026-09.02.2026	Data Visualization: Different types of plots	[1][2]	ICT	PPT
	12	11.02.2026	Plotting fundamentals using Matplotlib, Plotting fundamentals using Seaborn	[2]	ICT	PPT
	13	13.02.2026	Supervised Learning Techniques: Regression: Linear Regression, Multiple Linear Regression	[2]	ICT	PPT
	14	16.02.2026	Polynomial Regression	[1][2]	ICT	PPT
	15	18.02.2026	Logistic Regression	[1][2]	ICT	PPT

	16	18.02.2026	Regularization	[2]	Regular	NA
	17	18.02.2026	Ridge Regression, Lasso Regression	[2]	Regular	NA
	18	20.02.2026	Classification: Binary Classification and Multi-Class Classification,	[2]	Regular	NA
	19	20.02.2026	Support Vector Machine, K-Nearest Neighbours	[2]	Regular	NA
	20	23.02.2026-	Naive Bayes classifier,	[1][2]	Regular	NA
	21	23.02.2026-	Decision Trees	[2]	Regular	NA
	22	25.02.2026	Random Forest.	[2]	Regular	NA
C	23	27.02.2026-02.03.2026	Unsupervised Learning Techniques: Clustering: Centroid-Based Clustering	[1][2]	Regular	NA
	24	06.03.2026	K- Means Clustering	[1][2]	Regular	NA
	25	09.03.2026	Density-Based Clustering	[1][2]	Regular	NA
	26	11.03.2026-13.03.2026	DBSCAN Clustering Algorithm	[1][2]	Regular	NA
	27	16.03.2026	DBSCAN Clustering Algorithm Step	[1][2]	Regular	NA
	28	16.03.2026	DBSCAN Clustering Algorithm Feature	[1][2]	Regular	NA
	29	16.03.2026	DBSCAN Clustering Algorithm	[1][2]	Regular	NA
	30	16.03.2026	Distribution-Based Clustering	[1][2]	Regular	NA
	31	18.03.2026	Hierarchical Clustering.	[1][2]	Regular	NA
	32	18.03.2026	Association Rule Learning:	[1][2]	Regular	NA
	33	20.03.2026	Overview of the Association based	[1][2]	Regular	NA
	34	20.03.2026	Clustering and its Algorithms	[1][2]	Regular	NA
D	35	27.03.2026	Reinforcement Learning: Types of Reinforcement learning, Key Features of Reinforcement Learning	[1][2]	Regular	NA
	36	30.03.2026	Elements of Reinforcement	[1][2]	ICT	PPT
	37	1.04.2026-3.04.2026-06.04.2026	Learning, Applications of Reinforcement Learning	[1][2]	Regular	NA

38	8.04.2026- 10.04.2026- 13.04.2026	Performance Metrics: Performance Metrics for Regression: Mean Absolute Error, Mean Squared Error	[1][2]	ICT	PPT
39	15.04.2026-	Root Mean Squared Error	[1][2]	ICT	PPT
40	17.04.2026	R- Squared	[1][2]	ICT	PPT
41	17.04.2026	Performance Metrics for Classification	[1][2]	ICT	PPT
42	17.04.2026	Confusion Matrix	[1][2]	ICT	PPT
43	20.04.2026	Accuracy, Precision,	[1][2]	ICT	PPT
44	20.04.2026	Recall, F1 score.	[1][2]	ICT	PPT

Text Books:

1. Luke Welling and Laura Thomson, PHP and MySQL Web Development, Addison-Wesley Professional, 2013.
2. Josh Lockhart, Modern PHP: New Features and Good Practices, O'Reilly Media, 2015.
3. Robin Nixon, Learning PHP, MySQL & JavaScript: With jQuery, CSS & HTML5, O'Reilly Media, 2014.

Reference Books:

1. Mike McGrath, PHP 7 in Easy Steps, In Easy Steps Limited, 2016.
2. Larry Ullman, PHP for the Web: Visual QuickStart Guide, Peachpit Press, 2016.

Subject Teacher

Principal